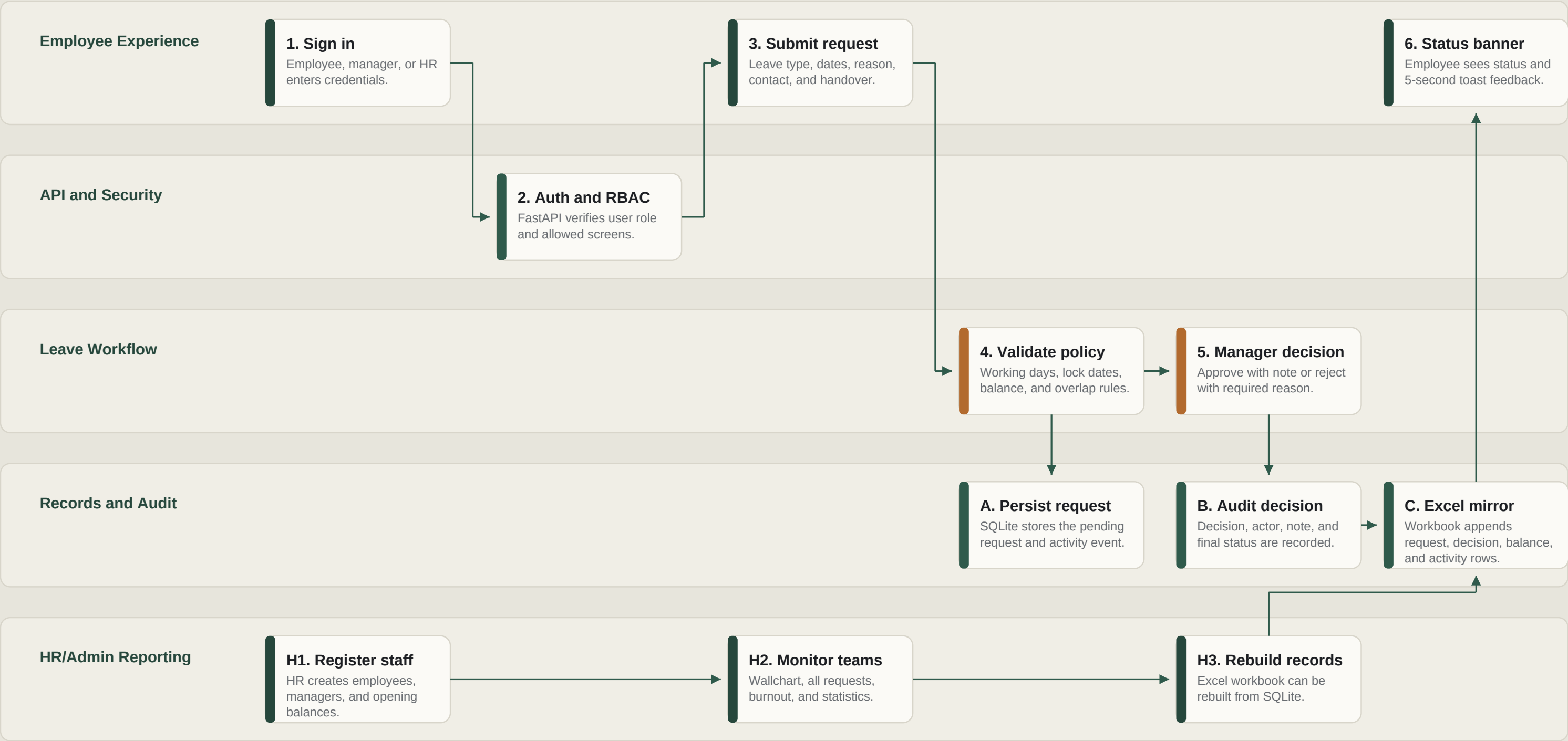


Industry-style leave management flow

A structured production workflow from role-based access through request validation, manager decision, audit persistence, user notification, and HR reporting.



Control points: authentication, RBAC, policy validation, manager accountability, SQLite audit trail, Excel operational mirror, HR reporting

Operational workflow responsibilities

This page explains the production-grade controls represented in the chart.

Access and session control

The backend authenticates users and exposes only role-appropriate data. Streamlit keeps the token only for the current session.

Employee request capture

The UI captures leave type, working dates, half-day state, reason, contact details, and handover notes before API submission.

Policy and balance validation

FastAPI calculates working days, checks locked dates and balance availability, and prevents unauthorized or invalid operations.

Manager accountability

Managers approve or reject only eligible direct-report requests. Rejections require a written reason for later review.

User-facing feedback

Submission and decision outcomes return to the UI as leave-type-specific 5-second floating banners.

SQLite source of truth

Employees, balances, requests, decisions, policy dates, activity events, and settings are stored in SQLite.

Excel operational mirror

The workbook records employees, balances, leave requests, decisions, policy dates, and activity for handoff and audit review.

HR/admin oversight

HR can register people, view all requests, monitor wallcharts, inspect burnout signals, and rebuild the workbook from SQLite.

Runtime file hygiene

Generated storage lives in `var/data/`. Source control keeps code, configuration, tests, docs, and `.gitkeep` placeholders.

Recommended hardening

Next production steps are signed auth tokens, Alembic migrations, CI checks, managed deployment, and durable external storage.